

Key Assumptions and Decisions

Database Engine Selection: Chose DuckDB as a local, embedded columnar database to execute sub-second analytical queries over millions of rows without heavy server infrastructure.

Star Schema Layout: Transitioned raw data lake files into a organized relational configuration of central facts and distinct dimension views to optimize analytical indexing.

Analytical Metric Conversion: Implemented regex string-stripping logic to clean currency punctuation from price fields so they could cast into functional numeric floats.

Scope Optimization Trim: Systematically removed planned sections for advanced data science, predictive pricing models, and text mining to prioritize core depth and output.

Geographic Target Focus: Emphasized solely on the London dataset to analyze data

Pricing Data missingness: Stripped all Null data in pricing for better data modelling

CSV Implementation to Database: Joined two main .csv files to query price, listing and date information that was null in .gz files.

Plotting graphs: Visualized data using different types of plot graphs according to use cases

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